

Due on 5th February 2026

Q1. (a) Consider an ideal gas containing N molecules, which is isothermally compressed from an initial volume V_i to a final volume V_f . Calculate the information content of the gas. Relate the result to the entropy variation in the gas. [2]

(b) Calculate the Boltzmann entropy of N spin $\frac{1}{2}$ particles subjected to a static magnetic field B_0 , at equilibrium temperature T . [4]

Q2. (a) Write down the truth table for the Toffoli Gate (a gate for reversible classical computation, also known as the controlled-controlled-NOT gate) which acts on three bits as follows: It flips the state of the third bit *iff* both the first and the second bits are in the state 1. [2]

(b) Suppose one represents the 8 possible states of the three bits by column vectors as follows:

$$000 \equiv \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}; 001 \equiv \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}; 010 \equiv \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}; 011 \equiv \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}; 100 \equiv \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}; 101 \equiv \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}; 110 \equiv \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}; 111 \equiv \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Construct the 8×8 matrix operator G which corresponds to the action of Toffoli Gate on the above vectors. [4]

(c) Show that $G^\dagger G = I$, where $G^\dagger$ is the adjoint operator of G (transpose of complex conjugate of G) and I is 8×8 the identity operator (this proves that G is a unitary operator, and thereby a valid quantum gate; use a clever method of identifying known matrices inside the gate before multiplying). [3]

Q3. (a) Express the Hadamard gate H as a product of the rotation matrices R_x and R_z and $e^{i\phi}$ for some phase ϕ . [3]

(b) An arbitrary single qubit unitary operator can be written in the form

$$U = e^{i\alpha} R_{\hat{n}}$$

for some real numbers α and θ , and a real 3-D vector $\hat{n}$.

(i) Prove this fact. (ii) Find the values of α , θ and $\hat{n}$ yielding the Hadamard gate H . (iii) Find the values of α , θ and $\hat{n}$ yielding the phase gate,

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$

[2+2+2]

Q4. Prove the following expression. Here, $\vec{\sigma} = \hat{i} \sigma_x + \hat{j} \sigma_y + \hat{k} \sigma_z$, and $\vec{a}$ and $\vec{b}$ are vectors.

$$(\vec{a} \cdot \vec{\sigma})(\vec{b} \cdot \vec{\sigma}) = (\vec{a} \cdot \vec{b}) \vec{I} + i \vec{\sigma} \cdot (\vec{a} \times \vec{b})$$

[3]